Course notes

Adding or multiplying

For pairwise disjoint finite sets A_i, $|\bigcup A_i| = \sum |A_i|$. If the cases overlap, the double counting must be corrected: $|A \cup B| = |A| + |B| - |A \cap B|$. The number of elements of the Cartesian product $A \times B$ is $|A||B|$. For a sequence of choices, multiply the numbers of possibilities at each stage, provided these numbers are valid for every earlier choice considered.

Example. A code made of two letters chosen from 26, followed by three digits, with repetition allowed, can take $26^2 \times 10^3$ values. A three-digit number does not begin with 0: there are 9×10^2 of them.

Lists and subsets

In a set of n elements, there are n^k ordered lists of length k with repetition; $n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}$ lists of k distinct elements, if $0 \leq k \leq n$; $n!$ permutations of all the elements. We set $0! = 1$.

Combinations

A combination of k elements from n is a subset with k elements; order does not matter.

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad \binom{n}{0} = \binom{n}{n} = 1, \quad \binom{n}{1} = n, \quad \binom{n}{2} = \frac{n(n-1)}{2}.$$

Why divide by $k!$? Each subset of size k gives exactly $k!$ distinct lists according to the order of its elements. The number of lists without repetition is therefore $k!\binom{n}{k}$.

Identities and proofs

$$\binom{n}{k} = \binom{n}{n-k}, \qquad \binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}, \qquad \sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

Pascal's rule above is used for $0 \leq k < n$; the boundary formulas are read directly from the values 1.

Symmetry. Taking the complement sets up a correspondence between the subsets of size k and those of size $n-k$.

Pascal, combinatorial proof. In a set of $n+1$ elements, fix an element a. A subset of size $k+1$ either contains a (choose k other elements from n) or does not contain it (choose $k+1$ elements from n). The two cases are disjoint and cover all the possibilities.

Pascal, algebraic proof. For $0 \leq k < n$, put over a common denominator:

$$\frac{n!}{k!(n-k)!} + \frac{n!}{(k+1)!(n-k-1)!} = \frac{n!((k+1)+(n-k))}{(k+1)!(n-k)!} = \frac{(n+1)!}{(k+1)!(n-k)!}.$$

The boundary cases are dealt with directly.

Number of subsets. For each of the n elements, choosing "present" or "absent" gives 2^n subsets. Sorting them by size, we count $\sum_{k=0}^{n} \binom{n}{k}$ of them: the two counts are equal.

Extension mentioned in the scheme of work. The expansion $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$ can be understood by choosing, in each of the n factors, the term a or b. This formula is not used as a prerequisite for the tests in this booklet.